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COHORT	CYBERSHUJAA COHORT 1-2026
TASK	MAL: MALWARE INTRODUCTORY REPORT

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MAL: MALWARE INTRODUCTORY REPORT

Introduction

In this module, I explored the fundamentals of malware analysis, including how malware campaigns operate and how infections can be identified. I learned about different analysis techniques such as static and dynamic analysis, as well as the tools used to investigate malicious files.

1.0 Understanding Malware Campaigns

In this section, I learned that malware campaigns are mainly classified into targeted and mass campaigns. Targeted attacks focus on specific individuals or organizations, while mass campaigns aim to infect as many devices as possible

Question: What is the famous example of a targeted attack-esque Malware that targeted Iran?

Answer: **Stuxnet**

Question: What is the name of the Ransomware that used the Eternalblue exploit in a "Mass Campaign" attack?

Answer: **Wannacry**

2.0 Identifying if a Malware Attack has Happened

In this section, I learned that malware attacks follow key stages such as delivery, execution, persistence, and propagation, each leaving behind traces of activity. I understood that malware can be identified through host-based and network-based signatures, such as file changes and unusual network traffic.

Question: Name the first essential step of a Malware Attack?

Answer: **Delivery**

Question: Now name the second essential step of a Malware Attack?

Answer: **Execution**

Question: What type of signature is used to classify remnants of infection on a host?

Answer: **Host-Based Signatures**

Question: What is the name of the other classification of signature used after a Malware attack?

Answer: **Network-Based Signatures**

3.0 Static Vs. Dynamic Analysis

I learned that malware can be analyzed in two main ways: static and dynamic. Static analysis inspects the code safely without running it, giving a quick overview, while dynamic analysis runs the malware to observe its behavior in real-time, though it is riskier.

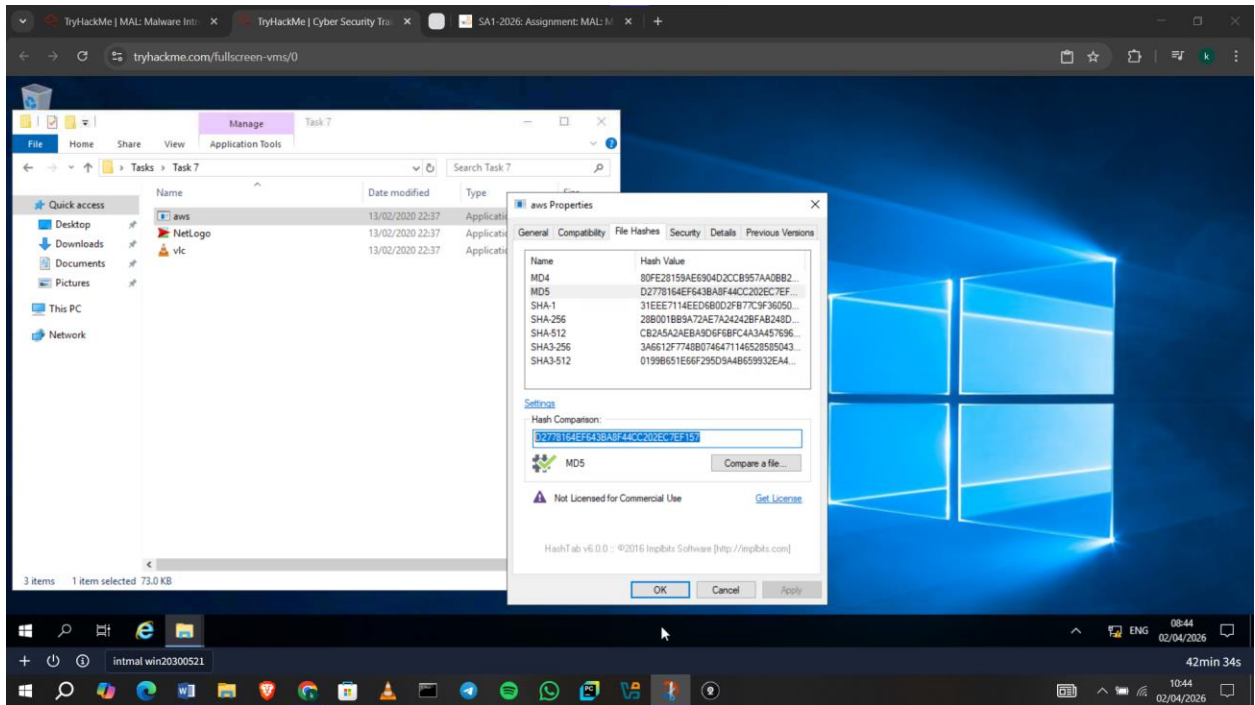
4.0 Obtaining MD5 Checksums of Provided Files

I learned that MD5 checksums act as unique digital fingerprints for files, allowing analysts to identify malware regardless of file name. I understood how tools like VirusTotal can use these checksums to quickly determine if a file has already been analyzed. This helps save time and improves accuracy when identifying suspicious files.

Question: The MD5 Checksum of aws.exe

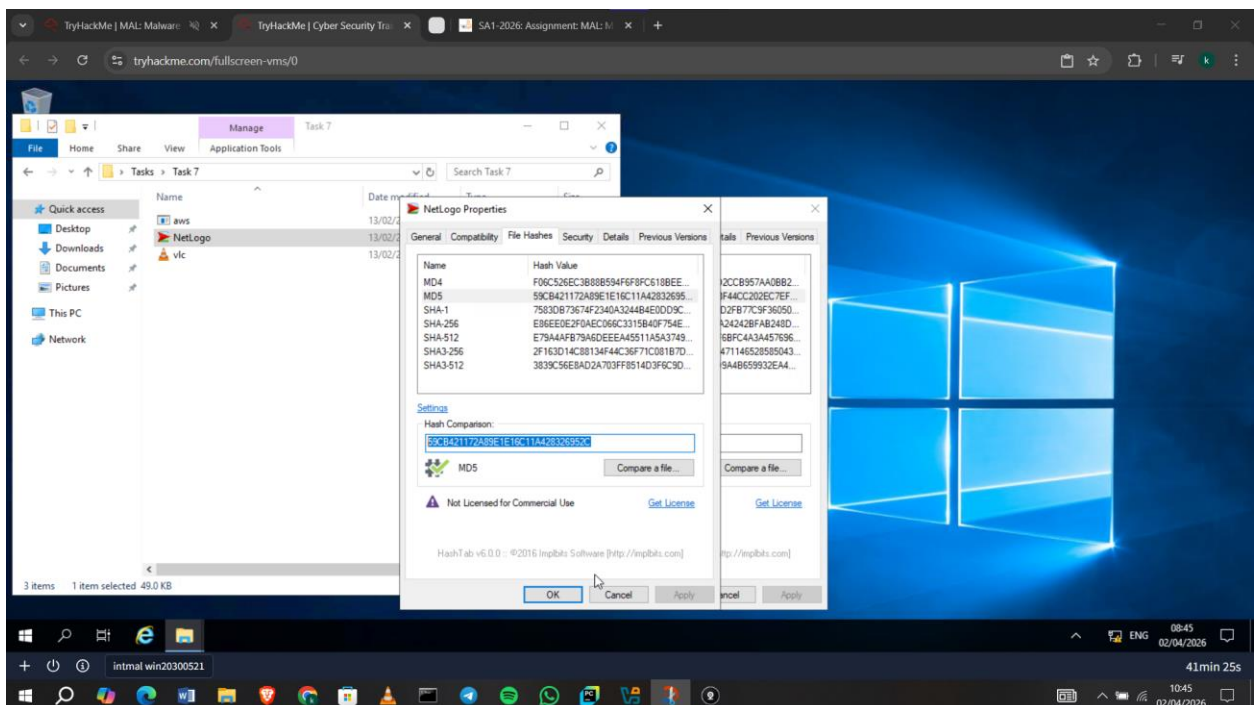
Answer: I opened the file properties and found it under **File Hashes:**

D2778164EF643BA8F44CC202EC7EF157



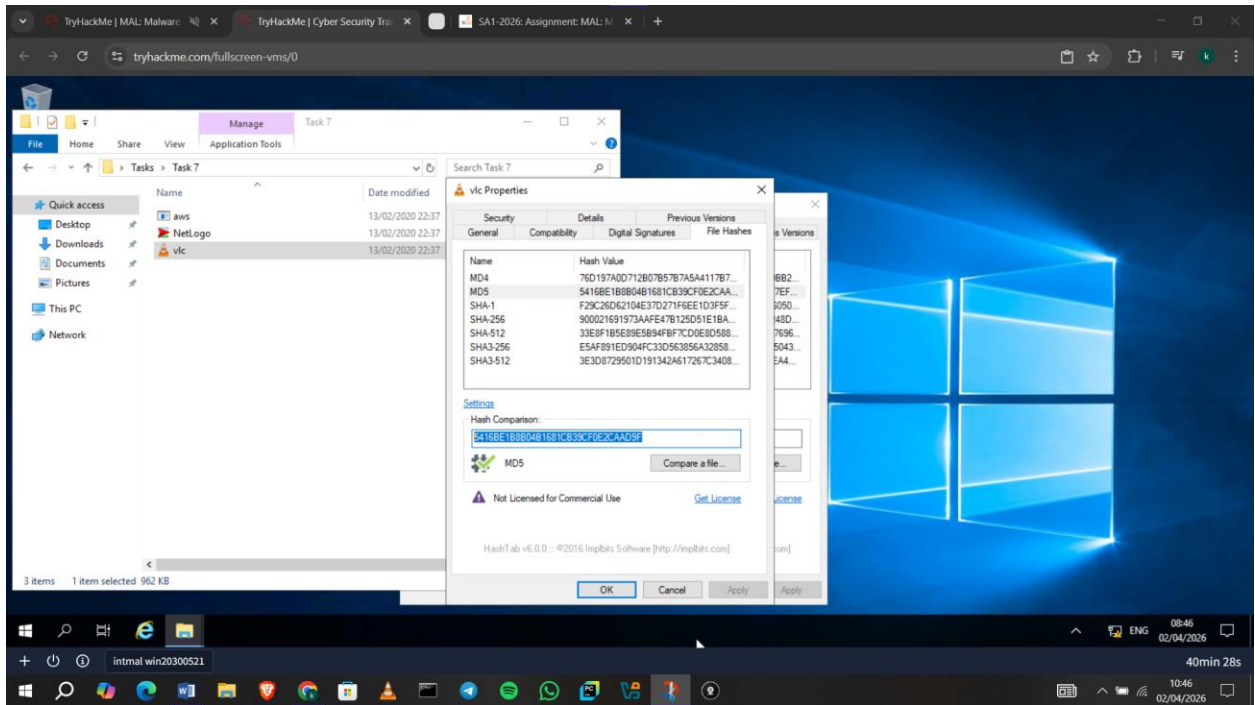
Question: The MD5 Checksum of Netlogo.exe

Answer: **59CB421172A89E1E16C11A428326952C**



Question: The MD5 Checksum of vcl.exe

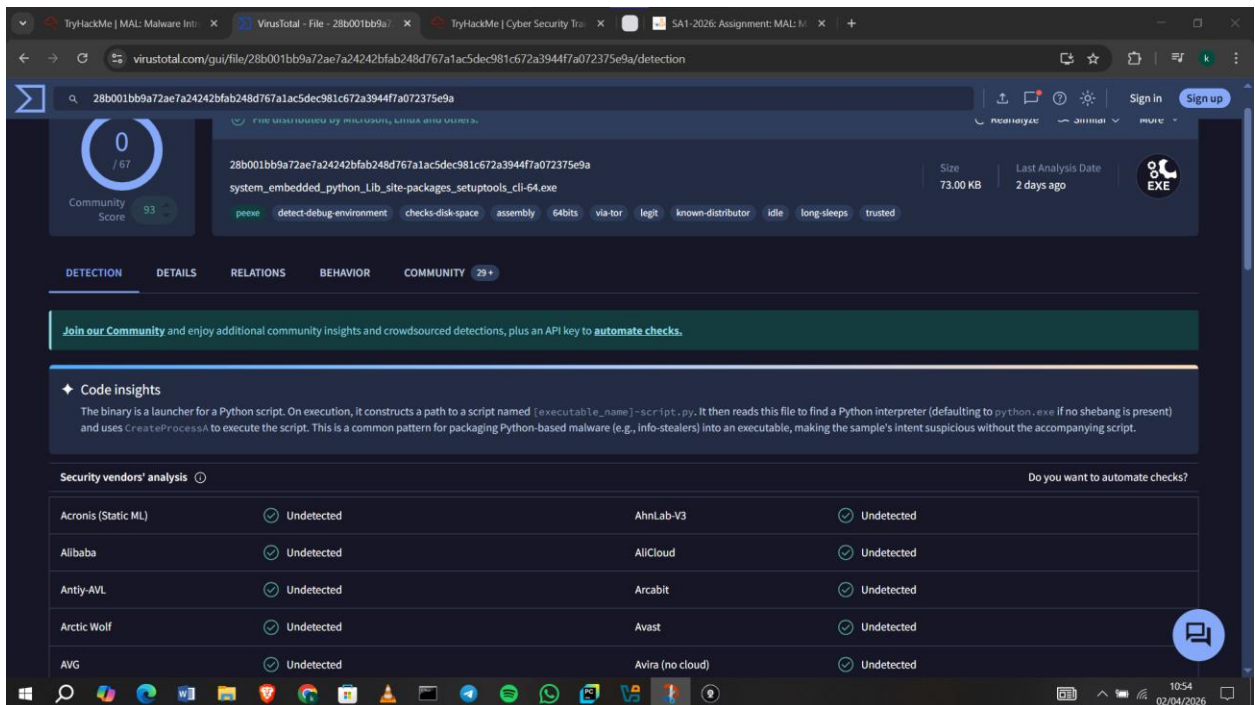
Answer: **5416BE1B8B04B1681CB39CF0E2CAAD9F**



5.0 MD5 Checksums Analysis check

Question: Does Virustotal report this MD5 Checksum / file aws.exe as malicious?

Answer: **Nay**. There were no vendor flags.



Question: Does Virustotal report this MD5 Checksum / file Netlogo.exe as malicious?

Answer: **Nay**

0 / 71
Community Score

No security vendors flagged this file as malicious

e86ee0e2f0aec066c3315b40f754ee25ac3c7d3db7dec20c2e82c8d9f5695536
NetLogo.exe
Size: 49.00 KB
Last Analysis Date: 6 days ago
EXE

peexe long-sleeps assembly detect-debug-environment via-tor 64bits

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Code insights
This binary is a loader component designed for a DLL side-loading attack. Its main function attempts to load packager.dll and execute the exported functions start_launcher and stop_launcher. These are non-standard exports for the legitimate Windows packager.dll, strongly indicating the program is intended to load a malicious payload masquerading as this system file. If the malicious DLL is not found, it displays a MessageBox with the error "packager.dll not found."

Security vendors' analysis
Do you want to automate checks?

Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
Alibaba	Undetected	AllCloud	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Arctic Wolf	Undetected	Avast	Undetected

Question: Does Virustotal report this MD5 Checksum / file vlc.exe as malicious?

Answer: **Nay**

0 / 72
Community Score

No security vendors flagged this file as malicious

900021691973aafe47b125d51e1bae5192760e91552dda0c7051226640c0a248
vlc.exe
Size: 962.70 KB
Last Analysis Date: 10 days ago
EXE

peexe detect-debug-environment overlay idle assembly 64bits long-sleeps

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 5

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis
Do you want to automate checks?

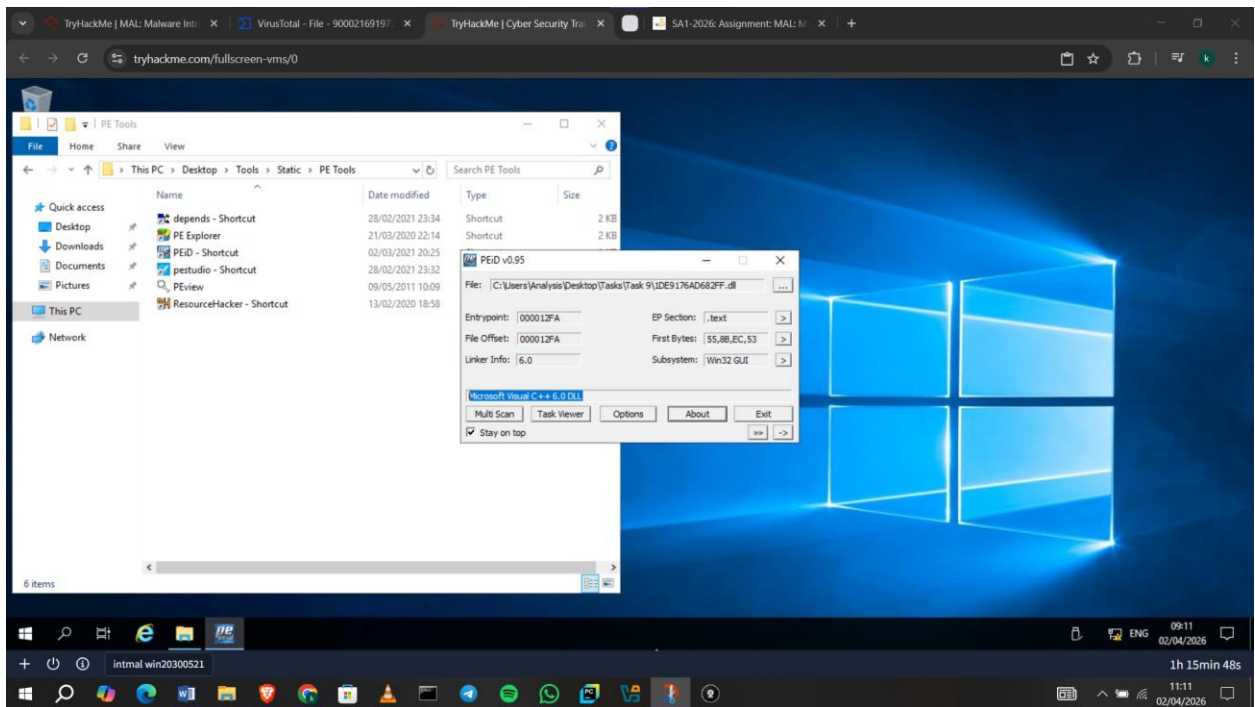
Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
Alibaba	Undetected	AllCloud	Undetected
ALYac	Undetected	Antiy-AVL	Undetected
Arcabit	Undetected	Arctic Wolf	Undetected
Avast	Undetected	AVG	Undetected
Avira (no cloud)	Undetected	Baidu	Undetected

6.0 Identifying if the Executables are obfuscated / packed

In this section, I learned that malware can be packed or obfuscated to hide its true behavior, making it harder to analyze. I understood how tools like PEiD can help identify the compiler or packer used in a file. I also learned that file extensions can be misleading, and that file headers such as “4D 5A” are more reliable for identifying executables.

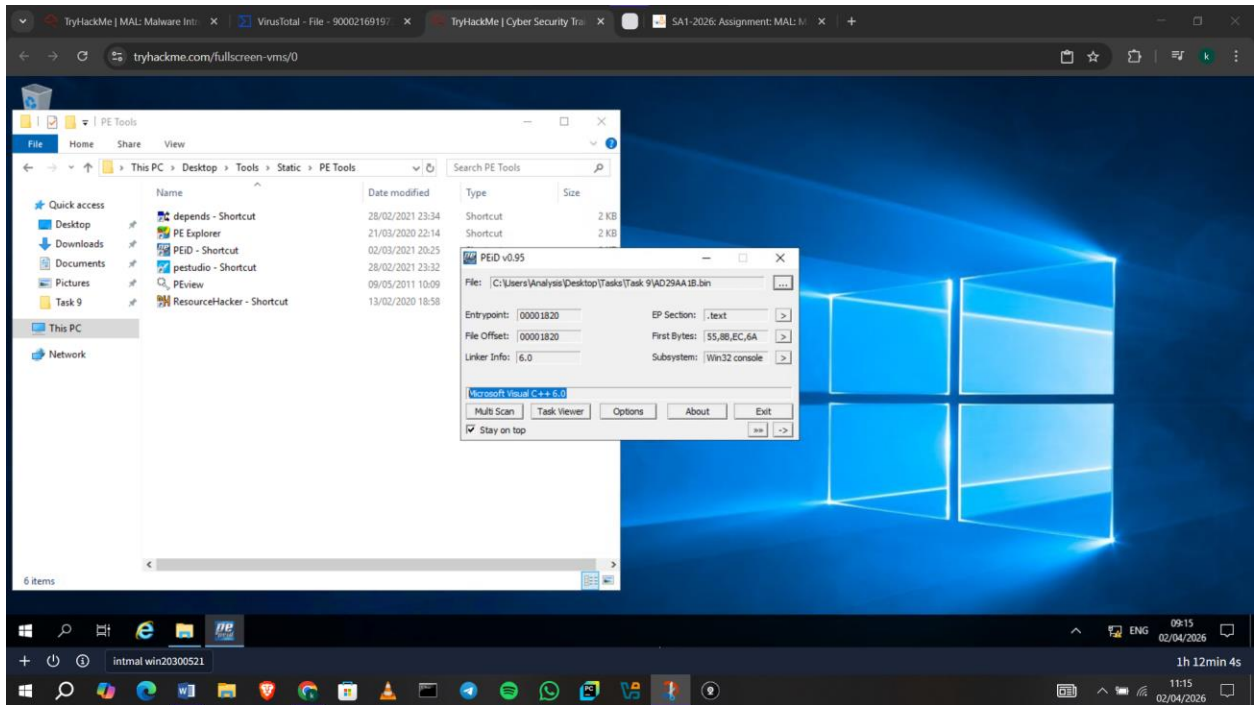
Question: What does PeiD propose 1DE9176AD682FF.dll being packed with?

Answer: **Microsoft Visual C++ 6.0 DLL**



Question: What does PeiD propose AD29AA1B.bin being packed with?

Answer: **Microsoft Visual C++ 6.0**

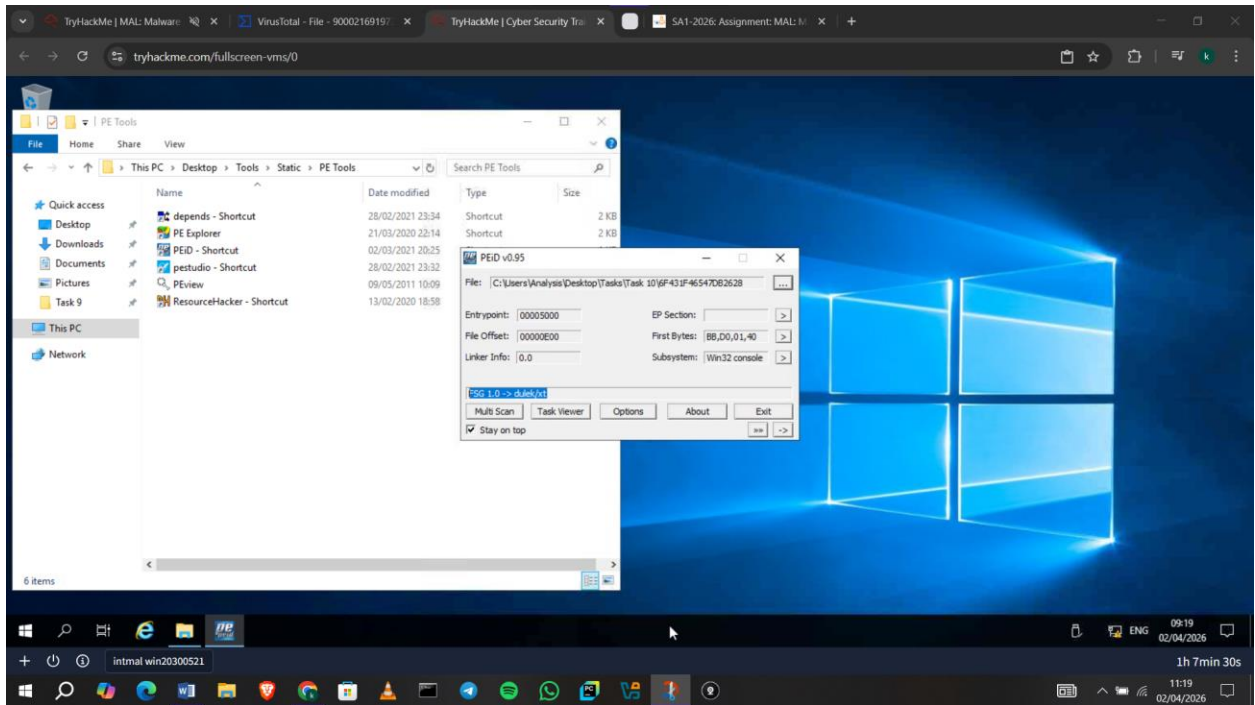


7.0 What is Obfuscation / Packing

I learned that obfuscation is used to hide the true functionality of a program, with packing being a common technique that compresses or encrypts code. While obfuscation can be used for legitimate purposes like protecting intellectual property, malware authors use it to avoid detection and make analysis more difficult.

Question: What packer does PeID report file "6F431F46547DB2628" to be packed with?

Answer: **FSG 1.0 -> dulek/xt**



8.0 Visualizing the Differences Between Packed & Non-Packed Code

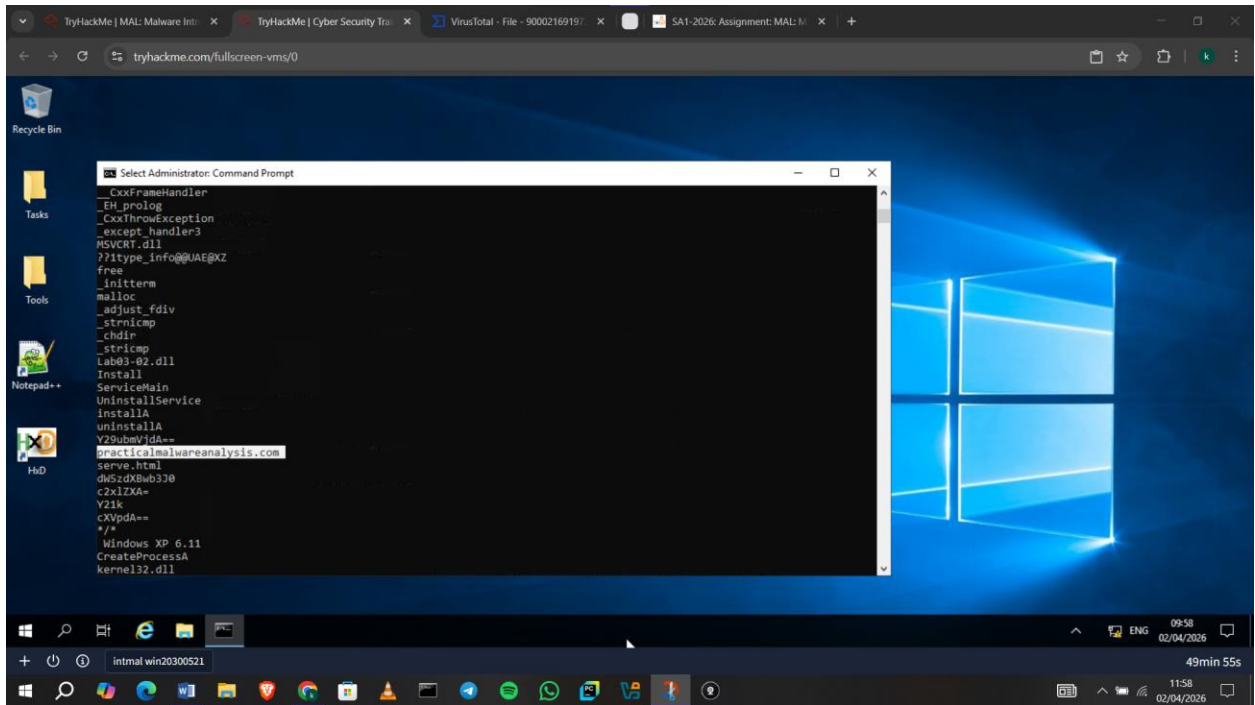
In this section, I learned how to visually distinguish between packed and non-packed executables using tools like PEiD and IDA Freeware. Packed files contain very little visible information, such as minimal imports and simple structures, making them harder to analyze.

9.0 Introduction to Strings

I learned that strings are readable text within a program that can reveal useful information about its behavior. I used tools like the Sysinternals strings utility and PE Explorer to extract and analyze these strings. I understood how analysts can identify potential malware activity without executing the file.

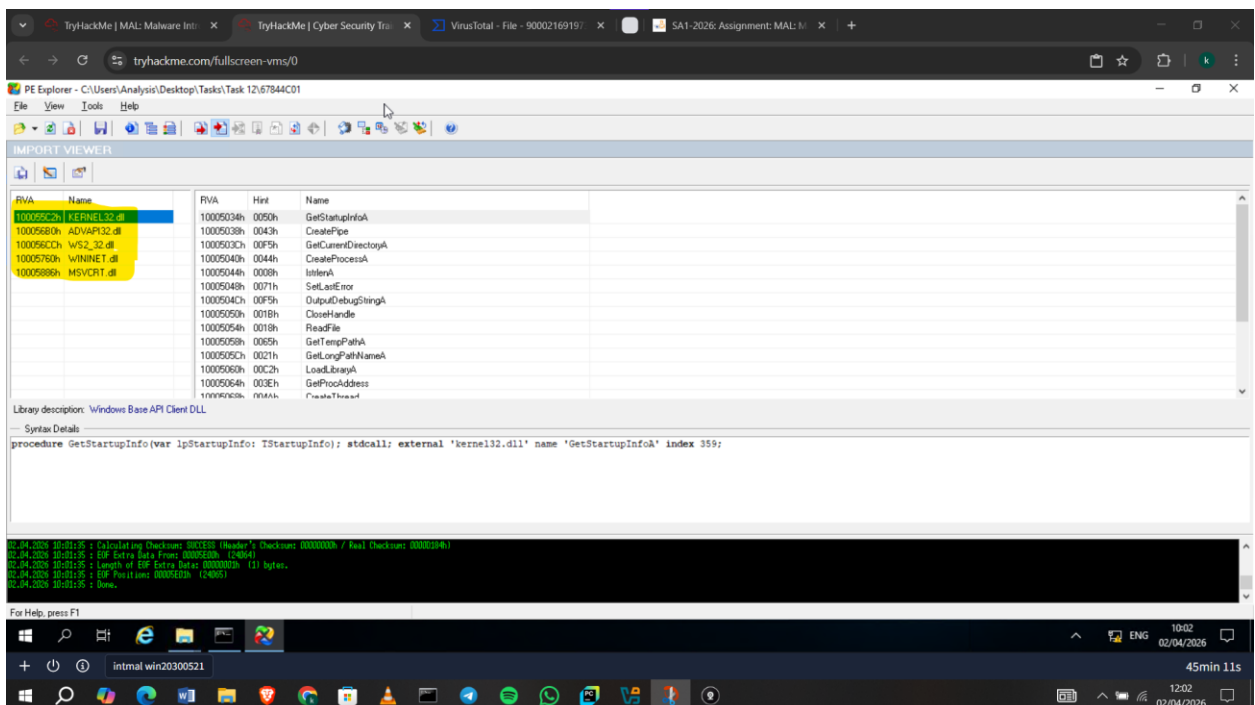
Question: What is the URL that is outputted after using "strings"

Answer: I navigated to **C:\Users\Analysis\Desktop\Tools\SysinternalsSuite** and then ran **strings "C:\Users\Analysis\Desktop\Task12\67844C01"** and found:
practicalmalwareanalysis.com



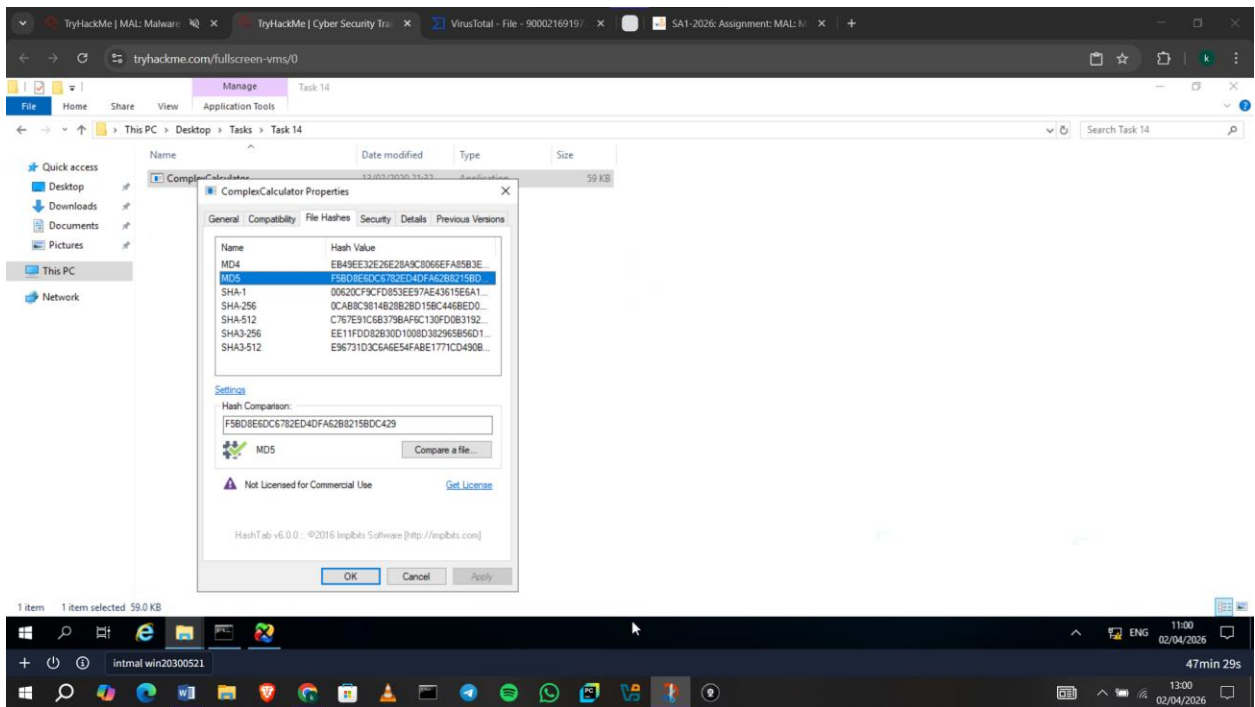
Question: How many unique "Imports" are there?

Answer: 5



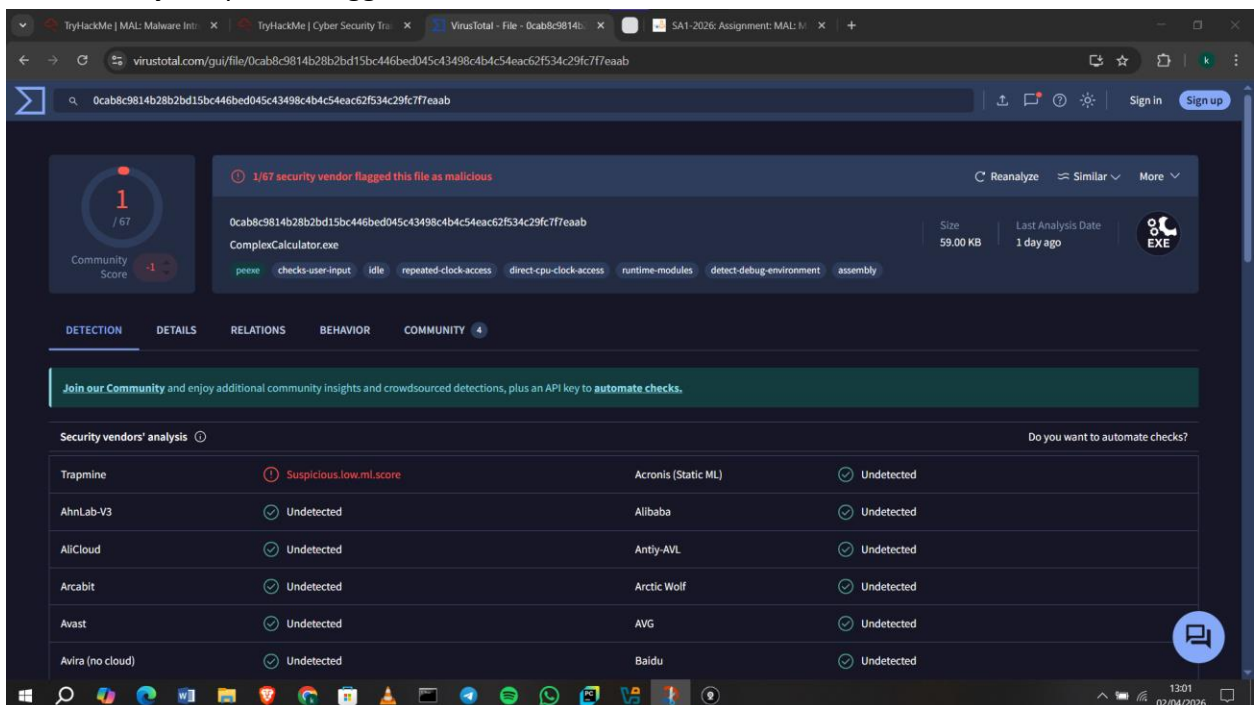
10.0 Introduction to Imports

Answer: f5bd8e6dc6782ed4dfa62b8215bdc429



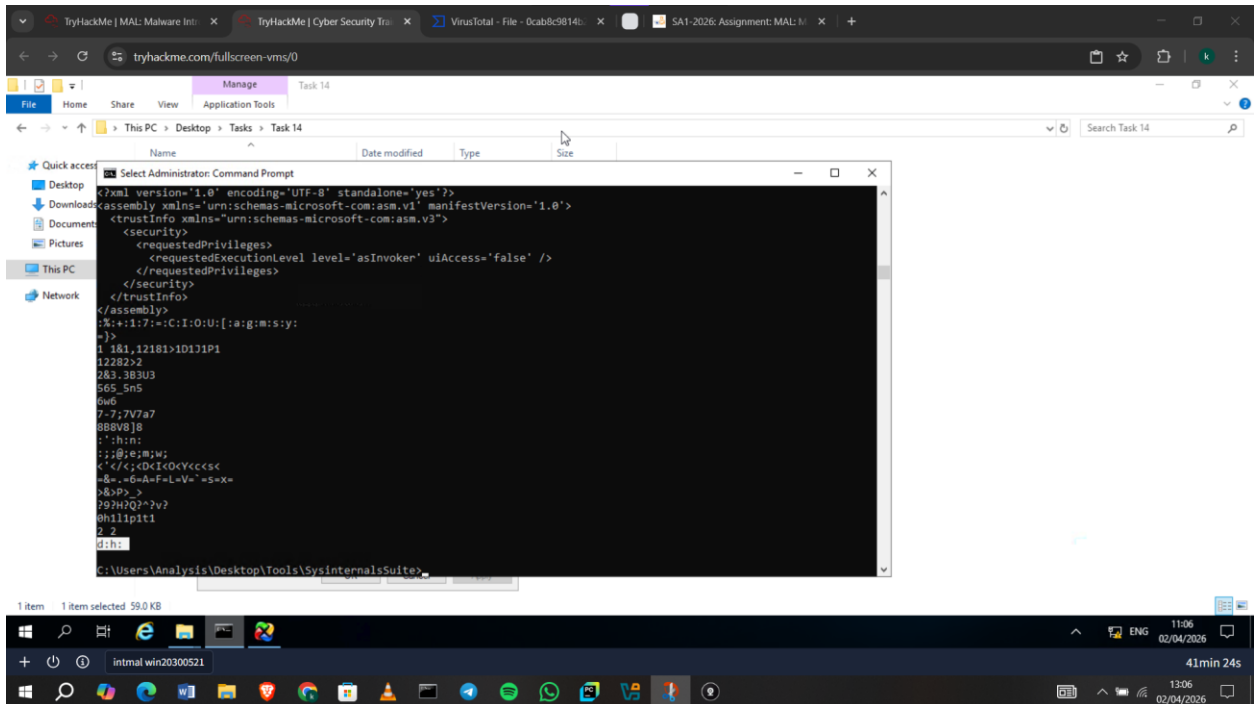
Question: Does Virustotal report this file as malicious?

Answer: **Yay.** Trapmine flagged it as malicious.



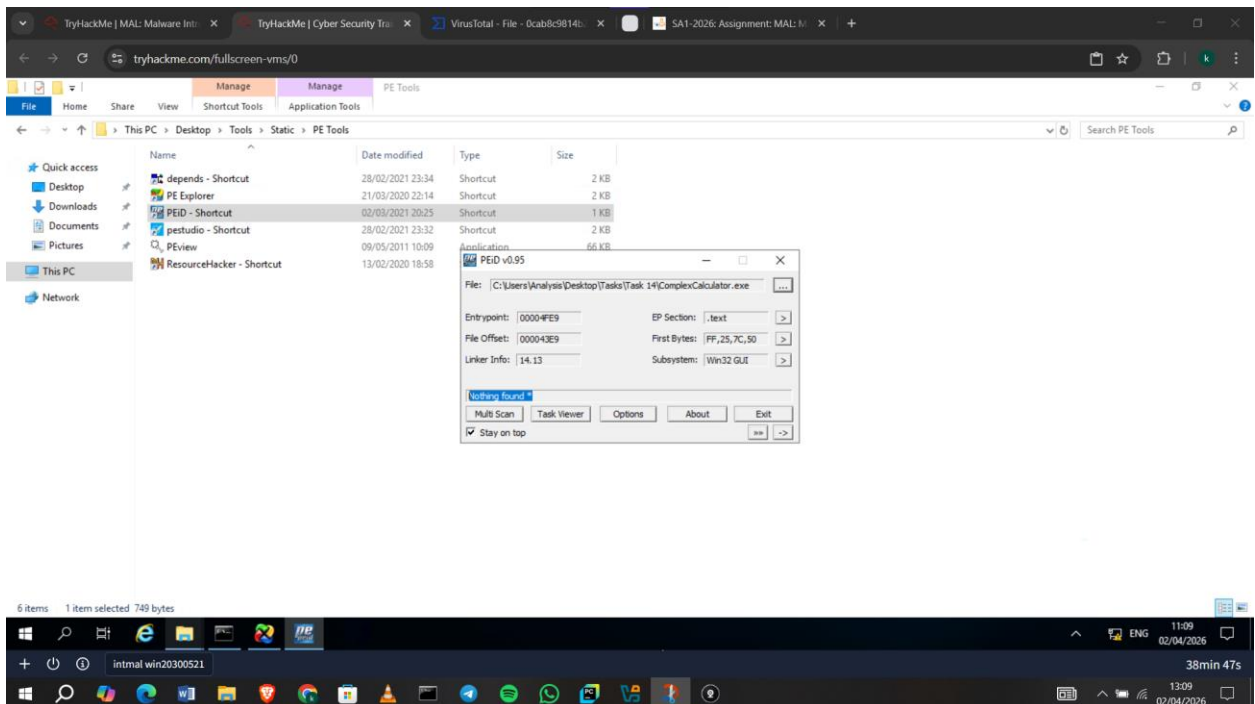
Question: Output the strings using Sysinternals "strings" tool. What is the last string outputted?

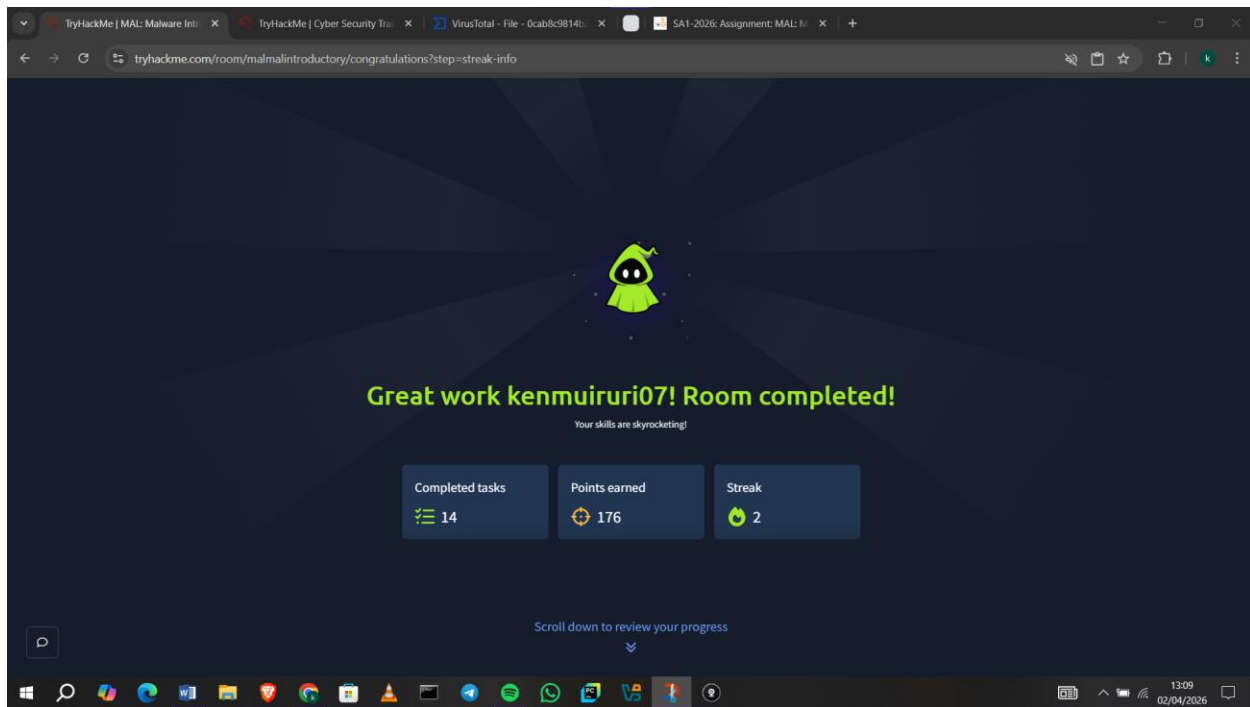
Answer: I ran **strings "C:\Users\Analysis\Desktop\Tasks\Task 14\ComplexCalculator.exe"** and found **d:h:**



Question: What is the output of PeID when trying to detect what packer is used by the file?

Answer: I got: **Nothing Found**





Conclusion

This module helped me understand how malware operates and the various methods used to analyze it. I gained practical knowledge on identifying malicious behavior through techniques like examining strings, imports, and file signatures.